Assignment - 1

Program - 1 - Design and implementation of an AND Gate using VHDL

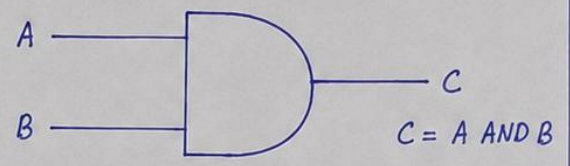
⇒ Inputs : A, B

⇒ Output : C

⇒ Truth table :

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

Circuit :

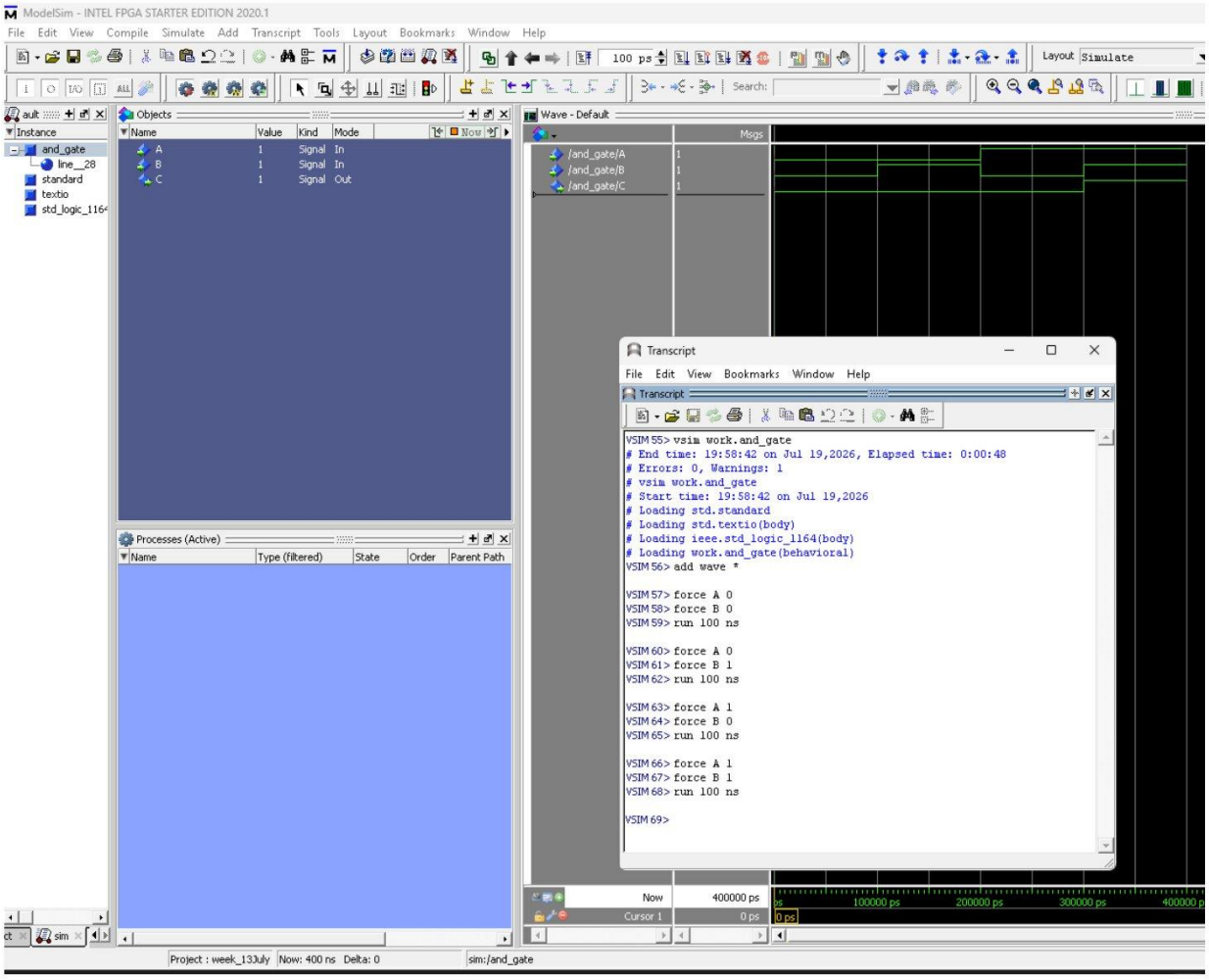


⇒ Program :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity and_gate is
    Port (
        A : in STD_LOGIC;   -- Input A
        B : in STD_LOGIC;   -- Input B
        C : out STD_LOGIC   -- Output C = A AND B
    );
end and_gate;

architecture Behavioral of and_gate is
begin
    -- Concurrent signal assignment:
    -- C gets the logical AND of A and B
       C <= A AND B;
end Behavioral;
```



ModelSim Simulation – AND Gate

Assignment - 1

Program - 1 - Design and implementation of an OR Gate using VHDL

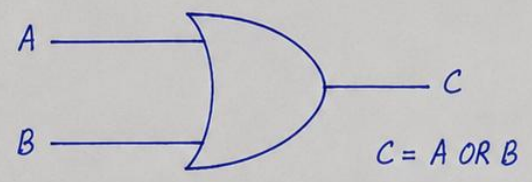
⇒ Inputs : A, B

⇒ Output : C

⇒ Truth table :

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

Circuit :

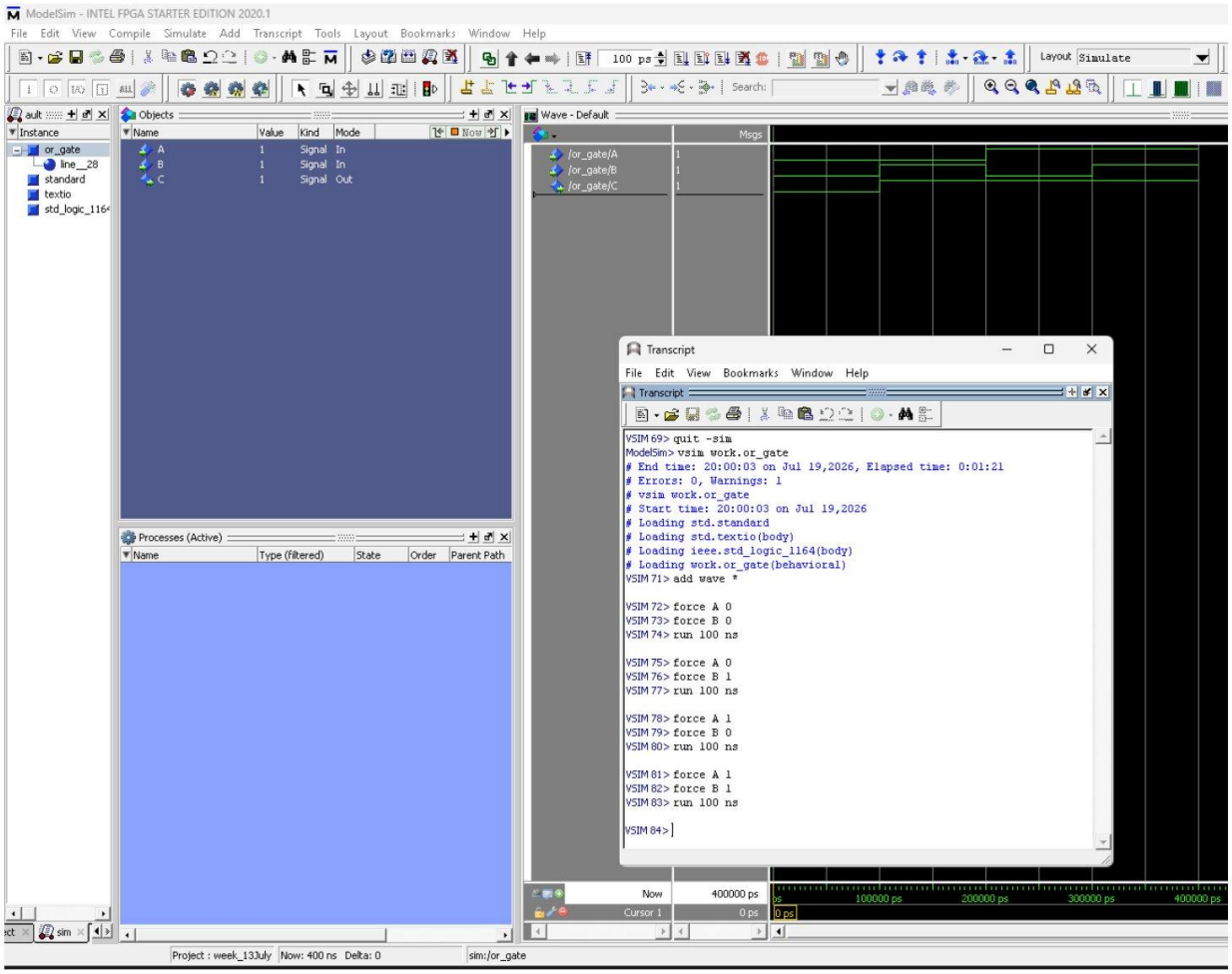


⇒ Program :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity or_gate is
    Port (
        A : in  STD_LOGIC;  -- Input A
        B : in  STD_LOGIC;  -- Input B
        C : out STD_LOGIC;  -- Output C = A OR B
    );
end or_gate;

architecture Behavioral of or_gate is
begin
    -- Concurrent signal assignment:
    -- C gets the logical OR of A and B
       C <= A OR B;
end Behavioral;
```



ModelSim Simulation – OR Gate

Assignment - 1

Program - 1 - Design and implementation of a NOT Gate using VHDL

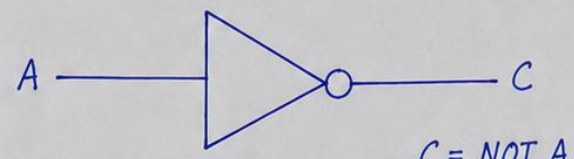
⇒ Inputs : A

⇒ Output : C

⇒ Truth table :

A	C
0	1
1	0

Circuit :



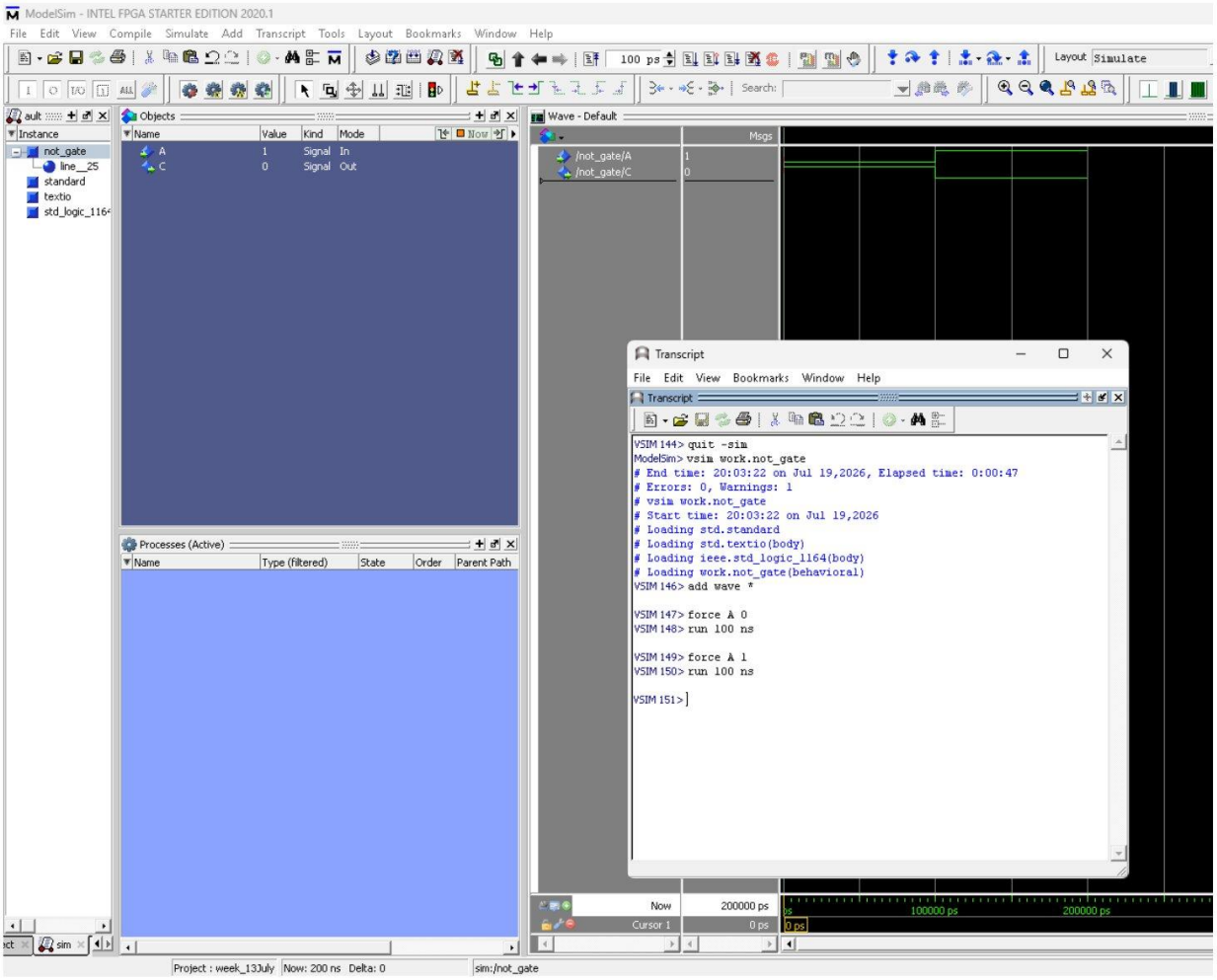
C = NOT A

⇒ Program :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity not_gate is
    Port (
        A : in  STD_LOGIC;   -- Input signal
        C : out STD_LOGIC    -- Output signal (inverted)
    );
end not_gate;

architecture Behavioral of not_gate is
begin
    -- Concurrent signal assignment:
    -- C gets the logical NOT of A
       C <= NOT A;
end Behavioral;
```



ModelSim Simulation – NOT Gate

Assignment - 1

Program - 1 - Design and implementation of a NAND Gate using VHDL

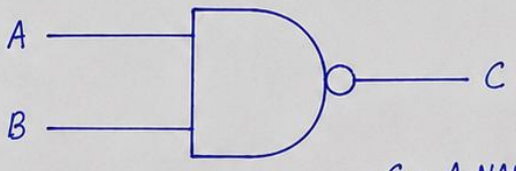
⇒ Inputs : A, B

⇒ Output : C

⇒ Truth table :

A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

Circuit :



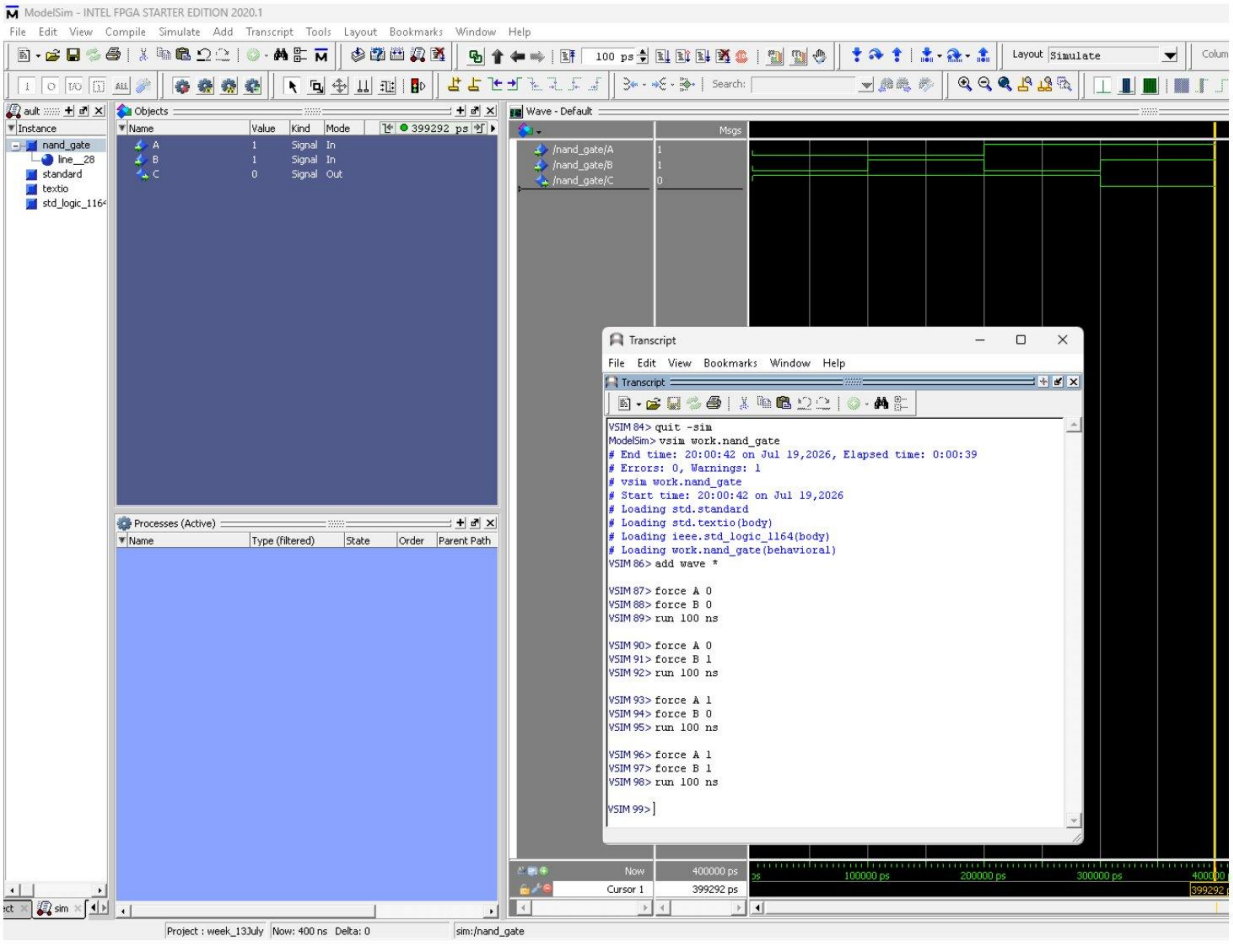
C = A NAND B

⇒ Program :

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity nand_gate is
    Port (
        A : in  STD_LOGIC;   -- Input A
        B : in  STD_LOGIC;   -- Input B
        C : out STD_LOGIC;   -- Output C = A NAND B
    );
end nand_gate;

architecture Behavioral of nand_gate is
begin
    -- Concurrent signal assignment:
    -- C gets the logical NAND of A and B
       C <= A NAND B;
end Behavioral;
```



ModelSim Simulation – NAND Gate

Assignment - 1

Program - 1 - Design and implementation of a NOR Gate using VHDL

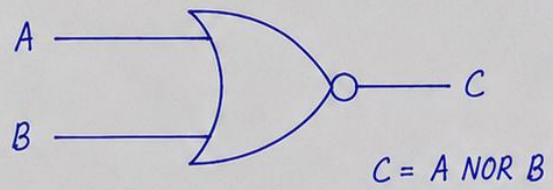
⇒ Inputs : A, B

⇒ Output : C

⇒ Truth table :

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

Circuit :



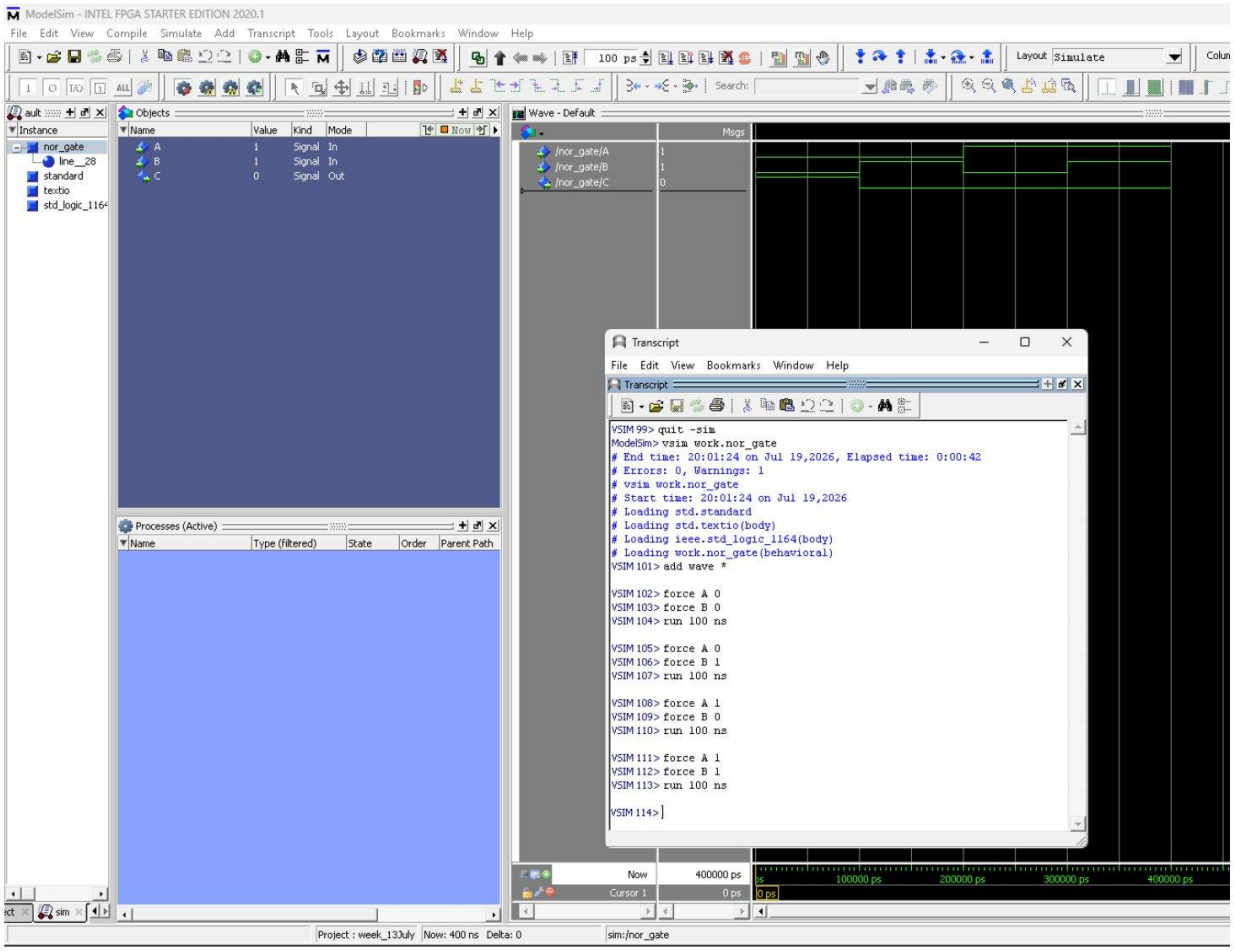
C = A NOR B

⇒ Program :

```
Library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity nor_gate is
    Port (
        A : in  STD_LOGIC;   -- Input A
        B : in  STD_LOGIC;   -- Input B
        C : out STD_LOGIC;   -- Output C = A NOR B
    );
end nor_gate;

architecture Behavioral of nor_gate is
begin
    -- Concurrent signal assignment:
    -- C gets the logical NOR of A and B
       C <= A NOR B;
end Behavioral;
```



ModelSim Simulation – NOR Gate

Assignment - 1

Program - 1 - Design and implementation of an XOR Gate using VHDL

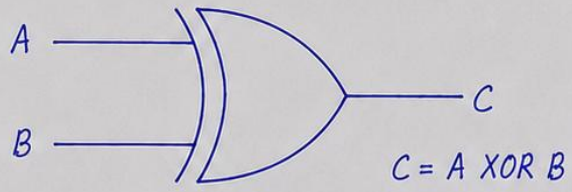
⇒ Inputs : A, B

⇒ Output : C

⇒ Truth table :

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

Circuit :



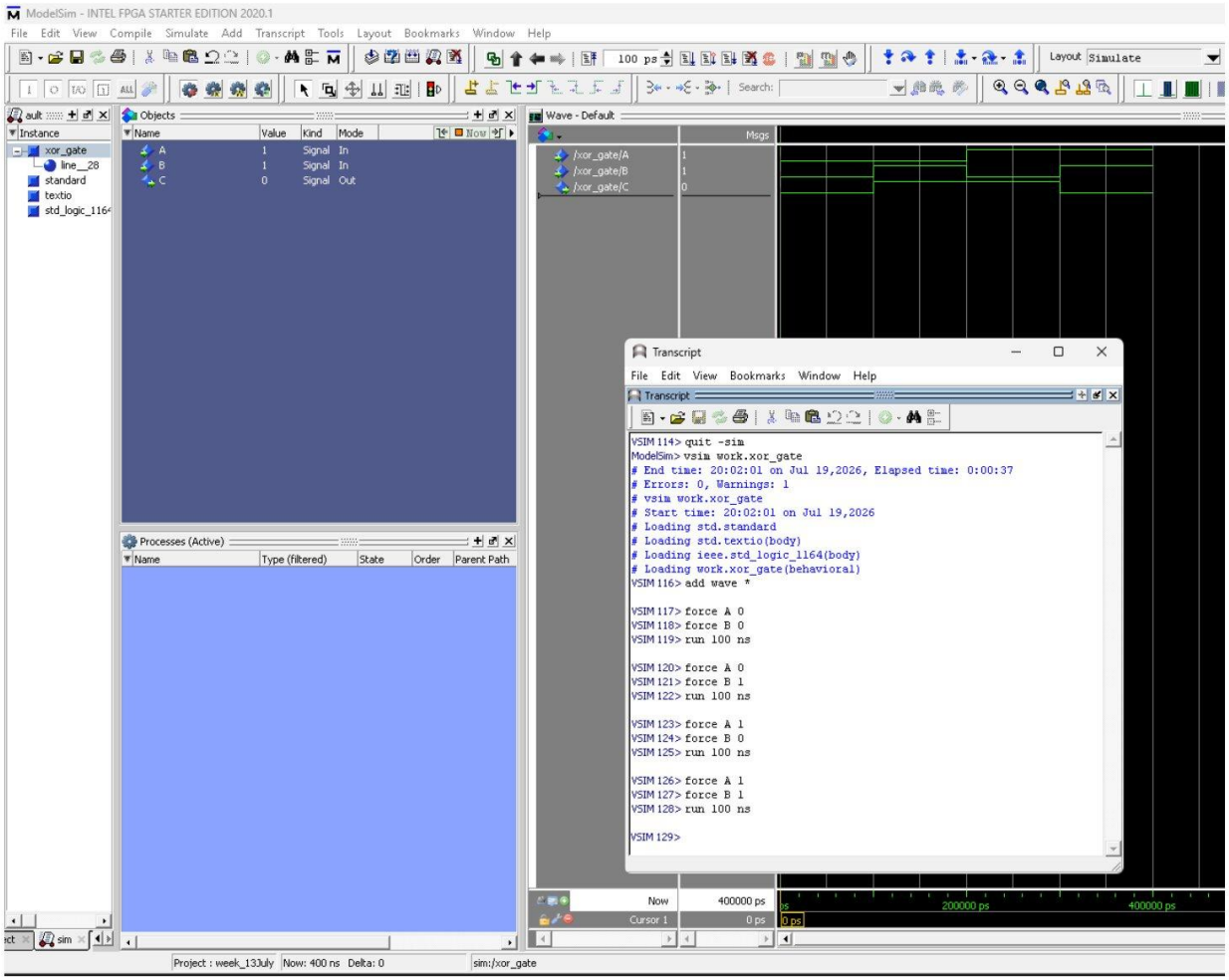
C = A XOR B

⇒ Program :

```
Library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity xor_gate is
    Port (
        A : in  STD_LOGIC;   -- Input A
        B : in  STD_LOGIC;   -- Input B
        C : out STD_LOGIC;   -- Output C = A XOR B
    );
end xor_gate;

architecture Behavioral of xor_gate is
begin
    -- Concurrent signal assignment:
    -- C gets the logical XOR of A and B
       C <= A XOR B;
end Behavioral;
```



ModelSim Simulation – XOR Gate

Assignment - 1

Program - 1 - Design and implementation of an XNOR Gate using VHDL

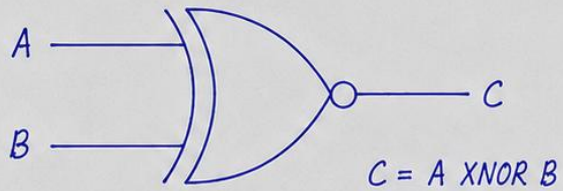
⇒ Inputs : A, B

⇒ Output : C

⇒ Truth table :

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

Circuit :



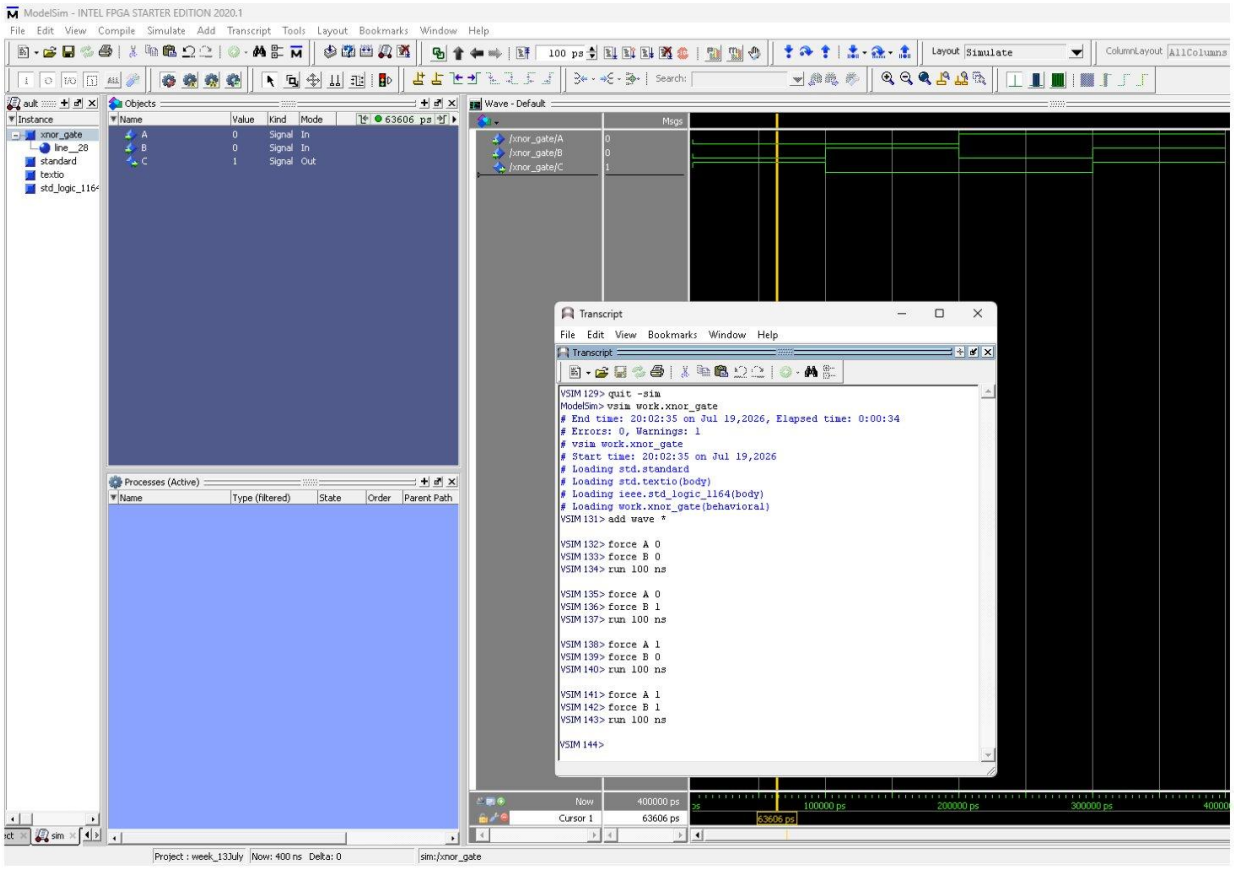
C = A XNOR B

⇒ Program :

```
Library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity xnor_gate is
    Port (
        A : in  STD_LOGIC;   -- Input A
        B : in  STD_LOGIC;   -- Input B
        C : out STD_LOGIC    -- Output C = A XNOR B
    );
end xnor_gate;

architecture Behavioral of xnor_gate is
begin
    -- Concurrent signal assignment:
    -- C gets the logical XNOR of A and B
       C <= A XNOR B;
end Behavioral;
```



ModelSim Simulation – XNOR Gate